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-- inbuilt functions:-
-- aggregate functions :-

-- string functions :-
-- 1) case conversion functions:-
-- a) upper()/ucase() :- converts string to uppercase
select upper(customer_name) from customers;
select ucase(customer_name) from customers;
-- b) lower()/lcase() :- converts string to lowercase
select lcase(customer_name) from customers;
select lower("VAISHVIK123@gmail.com");

-- 2) length functions
-- a) length() :- returns number of bytes
select length("vaishvik"); -- 8
select length("☑"); -- 3
select length("हिंदी वर्णमाला"); -- 40

-- b) char_length() :- return number of characters:_
select char_length("vaishvik"); -- 8
select char_length("♥"); -- 1
select char_length("हिंदी वर्णमाला"); -- 14

-- 3) concatenation functions
-- a) concat():- joins string
select concat("vaishvik", " ", "Singh");

create table temp_table(first_name varchar(225), last_name varchar(225));
insert into temp_table values("shree","Nath"),
("Rohan","singh"),("Paras","Sharma");

select *, concat(first_name," ",last_name) as full_name from temp_table;

alter table temp_table add column email varchar(225);
select * from temp_table;

update temp_table set email =
lcase(concat(first_name,last_name,"@gmail.com"));

-- b) Concat_ws() :- concatenate with separator:-
select concat_ws("-", "01", "01", "2025");

-- 4) Substring functions:-
-- a) substring()/substr():-
select substr("MySQL database",1,5);

-- b) left():-
select left("database",4);

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-- c) right()
select right("database",4);

-- 5) Searching functions:-
-- a) Instr():- returns position of first occurrence.alter
select
instr("database is a great technology. We should all learn database.",
'Popular');

-- locate():- similar to instr
select
locate('is',"database is a great technology. We should all learn database.
It a popular language",12);

-- c) position():- another way to search
select position('sql' in 'MySQL');

-- 6) Replace and modify:-
-- a) Replace:- Replaces a part of a string and it is case sensitive.
select replace("I love Java, Java is a really popular language","Java","MySQL");

select * from temp_table;

-- replace all employees email suffix with @yahoo.com where ever it's @gmail.com
select *, replace(email,"@gmail.com","@yahoo.com") from temp_table;
update temp_table set email = replace(email, "@gmail.com","@yahoo.com");

-- b) insert():- inserts text at specific position.
select insert("Database",5,0,"-");

-- 7) Trim functions:-
-- a) trim() :- removes leading and trailing spaces;
select trim("           MySQL           ");

-- b) ltrim():- removes left spaces
select ltrim("           MySQL           ");

-- c) rtrim() :- removes right spaces
select rtrim("           MySQL           ");

-- 8) padding functions:-
-- a) lpad():-
select first_name, last_name, char_length(lpad(first_name,10,"-")) from temp_table;
select lpad('vaishvik',10,'-');

-- b) rpad():-
select rpad('vaishvik',10,'*');

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-- 9) String comparison functions:-
-- a) strcmp():- compares two strings:-
-- return 0 if equal
-- returns 1 if first greater
-- return -1 if second greater.
select strcmp("abc", "abc");

-- 10) Reverse and repeat:-
-- a) reverse() :-
select reverse("mysql");

-- b) repeat():-
select repeat("sql",3);

-- 11) Format functions:-
-- format():-
-- format always return a string
select format(100000000000.378345,4);

-- ascii()
select ascii("%");

-- -----
-- numeric functions
-- Numeric functions works with numbers:-

-- 1) round():-
-- round is used to round off decimal numbers.
select round(34000.34564,2); -- 4.564 --> 5
select round(34000.34564);

-- 2) ceil()/ ceiling():- rounds up
select ceil(4.01); -- 5
select ceil(4.8); -- 5

-- 3) floor():- rounds down
select floor(4.99);
select floor(4.01);

-- 4) truncate():- truncate cuts decimal places without rounding.
select truncate(34.324935324,2); -- 34.32

-- 5) Power()/pow():-
select pow(2,3);
select pow(4,3);

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-- 6) sqrt():-
select sqrt(16);

-- 7) exp()
select exp(1);

-- 8) log():-
select log(10);

-- 9) log10():-
select log10(100);

-- 10) rand():- --> 0.01 --> 0.99
select rand();

select ceil(rand() * 100);

-- card game
select ceil(rand() * 52);

-- 11) greateast() and least():-

select greateast(45,23,63,24);
select least(45,23,63,24);

-- 12) div()
select 11 div 3;

-- -----
-- 3) Date and time functions:-
-- 1) now():- returns current date and time
select now();

create table test_table
(dob timestamp, dod datetime);

insert into test_table values(now(),now());
select * from test_table;

-- 2) curdate():- current_date():-
select curdate();

-- 3) curtime():-
select curtime();

-- 4) sysdate():-
select sysdate();

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-- 5) year():-
create table employees(emp_id int primary key auto_increment,
emp_name varchar(225), dob date);

insert into employees (emp_name,dob)
values("Shree","1980-03-23"),
("Khudeja","2004-04-23"),
("Priya","2005-05-23"),
("sonal","2010-05-12");

select * from employees;
alter table employees add column brith_year year;

update employees
set brith_year = year(dob);

select year(current_timestamp());
select year("2024-09-01 12:09:03");

-- month():_
select month("2024-09-01 12:09:03");

-- date():-
select date("2024-09-01 12:09:03");

-- day():-
select day("2024-09-22 12:09:03");

-- week():-
select week("2024-12-31 12:09:03");

-- hour():-
select hour("2024-09-01 12:09:03");

-- minute():-
select minute("2024-09-01 12:09:03");

-- second():-
select second("2024-09-01 12:09:03");

-- dayname():-
-- dayname is used to extract name of the day from date or datetime/tiimestamp
select dayname("2026-12-31 12:09:03");
select dayname(now());
select dayname("2004-08-5");
-- monthname()
select monthname("2004-08-5");

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-- date arithmetic :-
-- 1) date_add():-
-- date add is used to to add interval to any given date.
-- Interval can be in months, days, weeks, years
select date_add("2004-08-05", interval 10 month);

-- 2) date_sub():-
select date_sub("2004-08-05", interval 100 day);

-- adddate()/subdate():-
-- same as date_sub() and date_add():_

-- 3) datediff():-
-- datediff is used to find difference in days between two dates.
select datediff(now(),"2004-08-05");

select datediff(now(),"2024-12-12");
select datediff("2024-12-12",now());

-- 4) timestampdiff():-
-- timestampdiff can calculate difference in
-- year, month, day, hour, minute, second, week.
select timestampdiff(year,"2004-08-05", now());

select * from employees;
alter table employees add column months int;
update employees
set months = timestampdiff(month,dob,curdate());

-- format date:-
-- common format specifiers:-

/*format      meaning
   %d          Day
   %m          month
   %Y          4 digit year
   %H          hour
   %i          minutes
   %s          seconds
   %y          2 digit year
   %D          day with suffix
   %M          month name
*/
-- DD-MM-YYYY
-- YYYY-MM-DD

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SELECT * FROM EMPLOYEES;
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SELECT DATE_FORMAT(CURDATE(), '%D-%M-%Y');
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alter table employees add column date_of_birth varchar(225);
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update employees  
set date_of_birth = date_format(dob, '%D-%m-%Y');
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-- last_day():-  
select last_day(now());
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select curdate();
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select now();
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```
select current_timestamp();
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